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CausalMixGPD: Bayesian Nonparametric Modeling with Optional Heavy-Tail Augmentation and Causal Extensions

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Abstract

Heavy-tailed outcomes arise frequently in settings where rare extreme events drive risk, cost, or policy decisions. Parametric models often provide adequate descriptions of the distributional center but can mischaracterize extreme behavior, whereas fully nonparametric approaches lack principled mechanisms for tail extrapolation. This article introduces CausalMixGPD, an R package for Bayesian modeling of heavy-tailed outcomes that splices Dirichlet process mixture models for the bulk of the distribution with optional generalized Pareto tails within a unified inferential framework. The package accommodates both unconditional and covariate-dependent mixture specifications, performs posterior computation via Markov chain Monte Carlo through nimble, and extends naturally to arm-specific outcome models for causal inference. Posterior predictive summaries are provided for densities, quantiles, means, survival curves, and causal contrasts, with particular emphasis on extreme quantiles and other tail-sensitive functionals. The same modeling framework additionally supports predictor-dependent clustering under multiple kernel families. Taken together, CausalMixGPD furnishes a practical and extensible workflow for heavy-tailed density estimation, model-based clustering, and distributional causal reporting.

Keywords: Bayesian nonparametrics; Dirichlet Process Mixture; Generalized Pareto Distribution; Extreme Value Theory; Cluster Analysis; Causal Inference; R; nimble.

1. Introduction

Causal inference is conventionally summarized through average treatment effects, yet many substantive applications demand distributional comparisons between treatment regimes. In clinical trials, for instance, investigators may wish to characterize how a treatment alters outcomes across the entire outcome scale, including effects among individuals in the upper

tail where events are rare but consequences can be severe. This shift from mean-based to distribution-based causal questions ties the credibility of causal analyses directly to the quality of the underlying outcome distribution estimates under competing treatment assignments.

A growing R ecosystem supports Bayesian causal inference through flexible outcome modeling and posterior uncertainty quantification, including tree-based and model-averaging approaches such as bcf, bartCause, and BCEE (R Package Authors 2024a, 2025a, 2023a). While these tools are effective for mean and heterogeneous effect estimation, distributional reporting poses additional challenges: quantiles estimated independently can cross, and tail-focused contrasts require the full outcome distribution rather than a collection of marginal summaries. CausalMixGPD addresses these challenges by modeling outcome densities directly and deriving quantiles, means, survival curves, and tail summaries from a single posterior predictive object.

Dirichlet process mixtures (DPM) offer a versatile nonparametric framework for density estimation, as they can accommodate multimodality and latent heterogeneity without prespecifying the number of mixing components (Escobar and West 1995). Within the R ecosystem, Bayesian nonparametric mixture modeling is well supported by several packages based on Dirichlet process ideas. dirichletprocess provides a flexible interface for specifying DPMs and fitting them by posterior simulation, yielding inference for mixture weights, component labels, and predictive densities under user-chosen kernels (R Package Authors 2023b). DPpackage offers a broad suite of Bayesian semi- and nonparametric models, including DP-based formulations for marginal and conditional densities, regression functions, and model-based predictive distributions (Jara, Hanson, Quintana, Müller, and Rosner 2011). More recently, BNPmix emphasizes practical Bayesian nonparametric mixture approaches, supporting posterior inference for clustering structure and density/regression estimation under DP and Pitman–Yor-type mixture specifications (Corradin, Canale, and Nipoti 2021).

The clustering perspective is also central to the present package. Because DP mixtures induce random partitions through latent allocation variables, they naturally support model-based subgroup discovery in addition to density estimation. Bayesian cluster analysis is typically reported through label-invariant summaries such as pairwise co-clustering probabilities (Binder 1978) and representative partitions selected via Dahl's least-squares criterion (Dahl 2006). In regression settings, clustering itself can depend on predictors: dependent Dirichlet process constructions allow covariates to modulate mixture weights, component-specific parameters, or both (MacEachern 1999, 2000; De Iorio, Müller, Rosner, and MacEachern 2004; Quintana, Müller, Jara, and MacEachern 2022), and predictor-dependent stick-breaking schemes provide a practical route for covariate-modulated cluster prevalence (Ren, Du, Carin, and Dunson 2011). Existing software supports portions of this workflow but typically focuses on bulk density estimation and clustering under a single kernel family without explicit accommodation for heavy-tailed data. Consequently, applications requiring both latent subgroup discovery and tail extrapolation still necessitate additional modeling outside the standard Bayesian nonparametric mixture workflow.

Estimating quantities far into the tail demands special care when data become scarce, and fitting a single model to heavy-tailed data can distort the fit in the central region. Extreme value theory (EVT) provides a principled framework for tail modeling and extrapolation. EVT characterizes the asymptotic distribution of threshold exceedances, enabling estimation of tail probabilities and quantiles beyond the range of observed data (Balkema and de Haan 1974; Pickands 1975; Coles 2001). Bayesian implementations of EVT are available in R for

settings requiring posterior uncertainty quantification of tail quantities. evdbayes provides Bayesian inference for standard extreme-value models via posterior simulation, enabling posterior and posterior-predictive summaries for extreme-event quantities (R Package Authors 2025b). For peaks-over-threshold (POT) approaches, POT implements a focused collection of tools for GPD modeling and threshold-based tail analysis (R Package Authors 2024b). For extreme quantile estimation and rare-event tail summaries, extremefit provides a dedicated software implementation (Durrieu, Grama, Jaunatre, Pham, and Tricot 2018). These existing tools typically emphasize either flexible bulk modeling or dedicated tail analysis; applications requiring both predictor-dependent clustering and causal contrasts in the upper tail still oblige users to integrate multiple disparate workflows.

This article introduces CausalMixGPD, an R package that unifies Bayesian nonparametric modeling with EVT, clustering, and causal analysis in the presence of covariates. The package provides three simultaneous capabilities:

(i) **BNP density estimation with an optional GPD tail:** the package fits DPMs below a threshold and can splice a generalized Pareto tail above it for tail probabilities, extreme quantiles, and rare-event prediction.

(ii) **Predictor-dependent clustering:** dedicated clustering wrappers support dependence through weights, component parameters, or both, and report label-invariant summaries via posterior similarity matrices and Dahl representative partitions (Binder 1978; Dahl 2006).

(iii) **Causal estimation via posterior functionals:** treatment-specific outcome distributions are modeled directly, so that average, restricted-mean, and quantile treatment effects are computed from the fitted distributions in both randomized and observational studies, with optional propensity-score augmentation in the latter.

Taken together, CausalMixGPD offers a unified workflow for density estimation, prediction, clustering, and causal reporting when tail behavior is of primary concern. The package is implemented in R, with computationally intensive components delegated to nimble (R Package Authors 2025c). The remainder of this article is organized as follows. Section 2 introduces the spliced model and the reported causal estimands. Section 3 describes the package workflow and user interface. Sections 4 and 5 illustrate the clustering and causal workflows through data analyses, and Section 6 concludes with a discussion of limitations and future directions.

2. Background and Preliminaries

Let the outcome–covariate pair be defined by

$$(Y, X) \in \mathbb{R} \times \mathbb{R}^p. \tag{1}$$

For a fixed covariate value x, define

$$F(y \mid x) = \Pr\,(Y \le y \mid X = x), \qquad f(y \mid x) = \frac{\partial}{\partial y} F(y \mid x), \tag{2}$$

and

$$Q(\tau \mid x) = \inf\Big\{y \in \mathbb{R} :\, F(y \mid x) \ge \tau\Big\}, \qquad 0 < \tau < 1. \tag{3}$$

CausalMixGPD treats $F(\cdot \mid x)$ as the primary inferential object. The center of the distribution is modeled nonparametrically, and the upper tail can optionally be replaced by a generalized Pareto component when extreme-value extrapolation is required. All reported quantities—densities, survival curves, quantiles, means, and causal contrasts—are posterior functionals of this fitted distribution.

2.1. Generalized Pareto Distribution for Threshold Exceedances

The tail region is defined by values exceeding a high threshold. Extreme value theory provides a standard approximation for such exceedances. Let $u(x)$ denote a high threshold, potentially depending on x, and define the exceedance

$$Z = Y - u(x), \tag{4}$$

considered conditionally on $Y > u(x)$. The conditional excess distribution is

$$F_u(z \mid x) = \Pr\big(Y - u(x) \le z \mid Y > u(x), X = x\big), \qquad z \ge 0. \tag{5}$$

Under broad regularity conditions, $F_u(\cdot \mid x)$ is well-approximated by a generalized Pareto distribution (GPD) when $u(x)$ is sufficiently large (Balkema and de Haan 1974; Pickands 1975; Coles 2001).

For $y > u(x)$, the corresponding GPD cumulative distribution function is

$$F_{\mathrm{GPD}}(y \mid u(x), \sigma(x), \xi) = \begin{cases} 1 - \left(1 + \xi \dfrac{y - u(x)}{\sigma(x)}\right)^{-1/\xi}, & \xi \neq 0, \\ 1 - \exp\left(-\dfrac{y - u(x)}{\sigma(x)}\right), & \xi = 0, \end{cases} \tag{6}$$

with support $y \ge u(x)$ when $\xi \ge 0$, and $u(x) \le y \le u(x) - \sigma(x)/\xi$ when $\xi < 0$. The corresponding density function is

$$f_{\mathrm{GPD}}(y \mid u(x), \sigma(x), \xi) = \begin{cases} \dfrac{1}{\sigma(x)} \left(1 + \xi \dfrac{y - u(x)}{\sigma(x)}\right)^{-1/\xi - 1}, & \xi \neq 0, \\ \dfrac{1}{\sigma(x)} \exp\left(-\dfrac{y - u(x)}{\sigma(x)}\right), & \xi = 0. \end{cases} \tag{7}$$

The associated GPD quantile function takes the form

$$Q_{\mathrm{GPD}}(\tau \mid u(x), \sigma(x), \xi) = \begin{cases} u(x) - \sigma(x) \log(1 - \tau), & \xi = 0, \\ u(x) + \dfrac{\sigma(x)}{\xi}\Big((1 - \tau)^{-\xi} - 1\Big), & \xi \neq 0, \end{cases} \qquad 0 < \tau < 1. \tag{8}$$

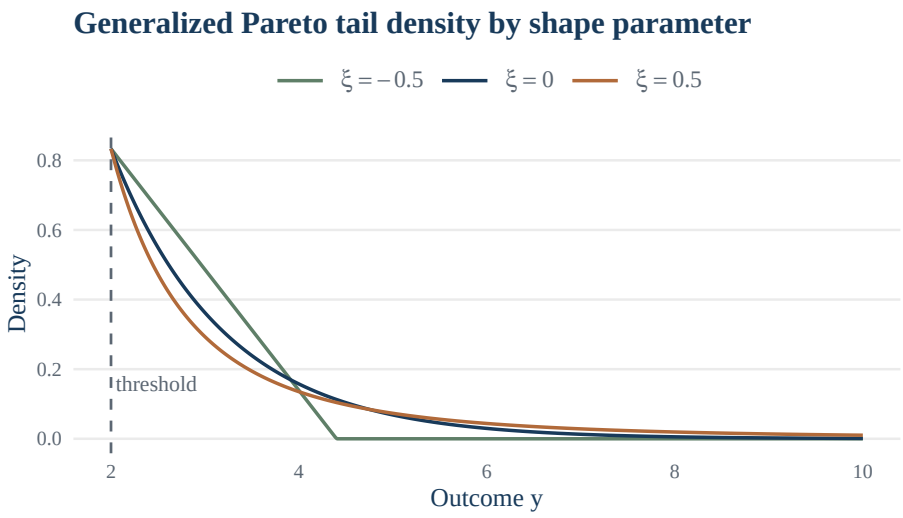


Figure 1: GPD tail density above threshold for multiple shape parameters.

Figure 1 illustrates the GPD density for several values of the shape parameter ξ with a fixed scale parameter σ and threshold u. The shape parameter governs tail heaviness: $\xi < 0$ yields a light tail with finite upper endpoint, $\xi = 0$ produces an exponential tail, and $\xi > 0$ generates a heavy tail with infinite support. This flexible family for modeling exceedances above a high threshold serves as the tail component in the spliced model described below.

2.2. Dirichlet Process Mixture Model for Conditional Densities

Below the threshold $u(x)$, the conditional density of Y given X is modeled using a DPM regression model. Let $k(\cdot \mid x; \theta)$ denote a user-chosen kernel family indexed by parameter θ. The hierarchical specification is

$$Y \mid X = x, \theta \sim k(\cdot \mid x; \theta), \qquad \theta \mid H \sim H, \qquad H \mid \kappa, H_0 \sim \mathrm{DP}(\kappa, H_0), \tag{9}$$

where H is an unknown mixing distribution on the kernel-parameter space, H_0 is a base measure, and $\kappa > 0$ is a concentration parameter (Ferguson 1973). Marginalizing over H yields an (almost surely) discrete mixing distribution together with an infinite mixture representation of the form

$$f_{\mathrm{DP}}(y \mid x) = \int k(y \mid x; \theta)\, dH(\theta) = \sum_{j=1}^{\infty} w_j\, k(y \mid x; \theta_j), \qquad w_j \geq 0, \qquad \sum_{j=1}^{\infty} w_j = 1. \tag{10}$$

One representation of the weights is Sethuraman's stick-breaking construction (Sethuraman 1994), in which

$$w_1 = V_1, \qquad w_j = V_j \prod_{\ell<j} (1 - V_\ell), \qquad V_j \overset{\text{iid}}{\sim} \mathrm{Beta}(1, \kappa), \qquad \theta_j \overset{\text{iid}}{\sim} H_0. \tag{11}$$

An equivalent characterization is obtained by marginalizing over H, yielding the Blackwell–MacQueen Pólya urn representation. In particular, the conditional distribution of a parameter

value given all others can be written as

$$\theta_i \mid \{\theta_j\}_{j\neq i} \sim \frac{\kappa}{\kappa+n} H_0(\cdot) + \frac{1}{\kappa+n}\sum_{j=1}^{n} \delta_{\theta_j}(\cdot), \tag{12}$$

where $\delta_{\theta_j}(\cdot)$ denotes a point mass at θ_j. This specification induces a clustering structure commonly described via the Chinese restaurant process (CRP) (Neal 2000). Both the stick-breaking and CRP representations are widely employed for posterior simulation in DPM models (Escobar and West 1995; Neal 2000). Because DPMs operate without prespecifying the number of mixture components and can capture complex distributional features such as multimodality, skewness, and heteroscedasticity through appropriate kernel choices, they constitute a natural model for the bulk of the conditional distribution where data are abundant and complex structure may be present.

2.3. Spliced DPM–GPD Conditional Distribution

CausalMixGPD employs a spliced (bulk–tail) construction that retains the DPM fit below a high threshold and replaces the upper tail with a GPD component (Frigessi, Haug, and Rue 2002; Behrens, Lopes, and Gamerman 2004; Coles 2001). For a fixed covariate value x, let $u(x)$ denote the threshold, $\sigma(x) > 0$ the tail scale above that threshold, and $\xi \in \mathbb{R}$ the tail-shape parameter. Let Θ collect the bulk-mixture parameters and Φ collect the tail quantities $\{u(x), \sigma(x), \xi\}$. Write $F_{\mathrm{DP}}(y \mid x; \Theta)$ for the bulk conditional CDF and $F_{\mathrm{GPD}}(y \mid x; \Phi)$ for the exceedance CDF. Then

$$p_u(x) = F_{\mathrm{DP}}(u(x) \mid x; \Theta) \tag{13}$$

is the fitted bulk mass below the threshold $u(x)$, and the spliced CDF is

$$F(y \mid x; \Theta, \Phi) = \begin{cases} F_{\mathrm{DP}}(y \mid x; \Theta), & y \leq u(x), \\ p_u(x;\Theta) + \{1 - p_u(x;\Theta)\}\, F_{\mathrm{GPD}}(y \mid x; \Phi), & y > u(x), \end{cases} \tag{14}$$

which is continuous at $u(x)$ by construction. Differentiating (14) yields

$$f(y \mid x; \Theta, \Phi) = \begin{cases} f_{\mathrm{DP}}(y \mid x; \Theta), & y \leq u(x), \\ \{1 - p_u(x;\Theta)\}\, f_{\mathrm{GPD}}(y \mid x; \Phi), & y > u(x), \end{cases} \tag{15}$$

where the tail density is rescaled by $1 - p_u(x;\Theta)$ to ensure that $f(\cdot \mid x; \Theta, \Phi)$ integrates to unity. Figure 2 provides a visual comparison of a one-component gamma density, a two-component gamma-mixture density, and a two-component gamma-mixture-plus-GPD spliced density with the splice point set at the 80th bulk-mixture quantile.

When the bulk kernel possesses a finite mean and the GPD tail satisfies $\xi < 1$, the ordinary mean of the spliced model admits the closed form

$$\mathrm{E}(Y \mid x; \Theta, \Phi) = \int_{-\infty}^{u(x)} y\, f_{\mathrm{DP}}(y \mid x; \Theta)\, dy \;+\; \{1 - p_u(x;\Theta)\}\left\{u(x) + \frac{\sigma(x)}{1-\xi}\right\}.$$

This identity underlies the analytical mean calculations employed by the package whenever the mean exists.

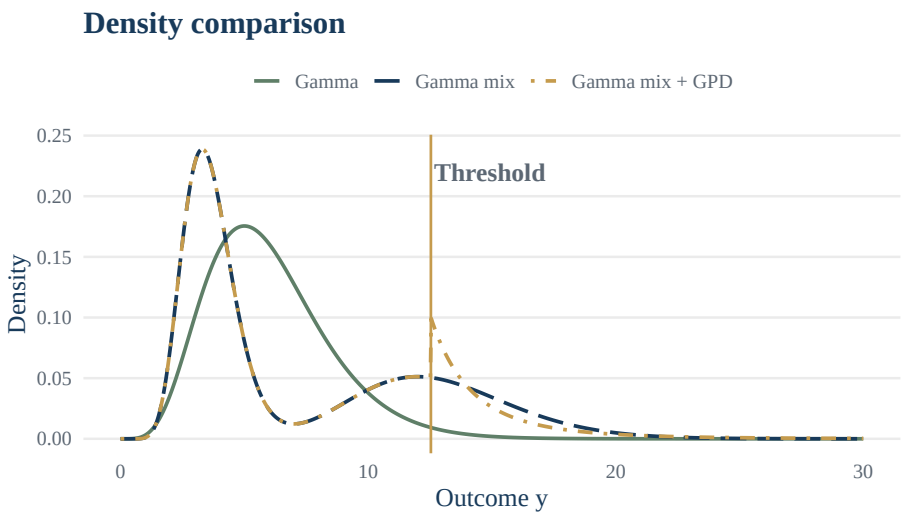


Figure 2: Comparison of three package-based curves: Gamma, Gamma mixture, and Gamma mixture + GPD with splice threshold set at the 80th percentile of the bulk mixture.

For $\tau \leq p_u(x;\Theta)$, quantiles are obtained by inverting the bulk CDF. Let $Q_{\mathrm{DP}}(\tau \mid x;\Theta)$ denote the bulk quantile function. For $\tau > p_u(x;\Theta)$, the probability index must be rescaled relative to the exceedance probability, yielding the rescaled index

$$\tilde{\tau}(x;\Theta) = \frac{\tau - p_u(x;\Theta)}{1 - p_u(x;\Theta)} \in (0,1), \tag{16}$$

Combining both cases, the spliced quantile function is

$$Q(\tau \mid x;\Theta,\Phi) = \begin{cases} Q_{\mathrm{DP}}(\tau \mid x;\Theta), & \tau \leq p_u(x;\Theta), \\ Q_{\mathrm{GPD}}\big(\tilde{\tau}(x;\Theta) \mid x;\Phi\big), & \tau > p_u(x;\Theta). \end{cases} \tag{17}$$

In subsequent sections, Θ indexes the bulk DPM and Φ indexes the tail component; their respective specifications appear in Sections 2.1 and 2.2.

2.4. Clustering under predictor-dependent Bayesian nonparametric mixtures

A standard Bayesian nonparametric approach to conditional density modeling represents the collection $\{f(\cdot \mid x) : x \in \mathcal{X}\}$ through predictor-indexed random probability measures. Within the Dependent Dirichlet Process (DDP) framework of MacEachern (1999, 2000), one specifies a family $\{G_x : x \in \mathcal{X}\}$ of random mixing measures whose dependence across x induces flexible borrowing of information in $f(\cdot \mid x)$.

A generic mixture representation is

$$f(y \mid x) \;=\; \int k(y \mid x,\theta)\; dG_x(\theta), \tag{18}$$

where $k(\cdot \mid x,\theta)$ is a kernel and G_x is a predictor-indexed mixing measure. When G_x is discrete (as under stick-breaking constructions), latent parameters exhibit ties, inducing a

random partition of observations. Introducing latent cluster-label variables z_i and component parameters $\{\theta_j\}_{j\geq 1}$ such that

$$y_i \mid z_i, \{\theta_j\}, x_i \sim k(\cdot \mid x_i, \theta_{z_i}), \tag{19}$$

clustering is characterized by the posterior distribution of the induced partition $\Pi = \{z_1, \ldots, z_n\}$ and is summarized in a label-invariant manner through co-clustering functionals (Binder 1978; Dahl 2006). The clustering component of CausalMixGPD accommodates three forms of predictor dependence, each corresponding to a particular construction of G_x.

- **Weight dependence only:** Predictors modulate mixture weights while component parameters remain shared across covariates,

$$f(y \mid x) \;=\; \sum_{j=1}^{\infty} w_j(x)\, k(y \mid \theta_j)\,. \tag{20}$$

  A common implementation uses predictor-dependent stick-breaking, where stick lengths are transformed regressions in covariates; the logistic stick-breaking process constructs these lengths via logistic regressions and maps them to weights (Ren *et al.* 2011). Such models can be viewed as practical DDP instances in which dependence is introduced through smoothly varying weights (MacEachern 2000; Quintana *et al.* 2022).

- **Parameter dependence only:** Predictors enter through component-specific kernel parameters while weights remain global,

$$f(y \mid x) \;=\; \sum_{j=1}^{\infty} w_j\, k(y \mid x, \theta_j)\,. \tag{21}$$

  This structure aligns with DDP constructions that share weights while permitting predictor-indexed parameters (MacEachern 1999, 2000), and is closely related to ANOVA-type dependent priors for structured collections of random measures (De Iorio *et al.* 2004).

- **Weight and parameter dependence:** The most flexible specification allows predictors to influence both cluster prevalence and within-cluster behavior,

$$f(y \mid x) \;=\; \sum_{j=1}^{\infty} w_j(x)\, k(y \mid x, \theta_j)\,. \tag{22}$$

  This corresponds to DDP-type models in which both selection probabilities and component-specific conditional structure vary with predictors (Quintana *et al.* 2022).

Posterior clustering summaries are reported through label-invariant functionals. The posterior similarity matrix (PSM) has entries

$$S_{i\ell} \;=\; \Pr(z_i = z_\ell \mid \text{data}), \qquad i, \ell = 1, \ldots, n, \tag{23}$$

estimated by Monte Carlo averages over MCMC output. A representative partition is obtained by selecting the sampled partition whose adjacency matrix is closest to the PSM under squared

loss (Dahl 2006), yielding a stable point estimate while retaining the PSM as the primary uncertainty summary. Section 3.3 provides further details on the package's implementation of these summaries.

2.5. Causal Inference: Notation, Identification, and Reported Estimands

We adopt the potential outcomes framework (Neyman 1990; Rubin 1974; Imbens and Rubin 2015). Let $A \in \{0, 1\}$ denote a binary treatment indicator ($A = 1$ treated, $A = 0$ control), let X denote pre-treatment covariates, and let $Y(a)$ denote the potential outcome under arm $a \in \{0, 1\}$. The observed outcome satisfies

$$Y = A\,Y(1) + (1 - A)\,Y(0). \tag{24}$$

The treatment-specific conditional distribution functions are the fundamental inferential objects. For each arm $a \in \{0, 1\}$, the arm-specific conditional CDF is

$$F_a(y \mid x) = \Pr\{Y(a) \le y \mid X = x\}, \qquad a \in \{0, 1\}. \tag{25}$$

In CausalMixGPD, the corresponding conditional densities $f_a(y \mid x)$ assume the spliced DPM structure in (15). The conditional τ-quantile is obtained by inverting $F_a(\cdot \mid x)$,

$$Q_a(\tau \mid x) = \inf\{y : F_a(y \mid x) \ge \tau\}, \qquad \tau \in (0, 1), \tag{26}$$

and the conditional mean is a functional of the conditional density,

$$\mu_a(x) = \mathrm{E}\{Y(a) \mid X = x\} = \int y\, f_a(y \mid x)\, dy. \tag{27}$$

Marginal (standardized) distributions are obtained by averaging conditional distributions over a covariate distribution. Writing F_a^{m} for the population-standardized arm-a CDF and F_a^{t} for the treated-standardized analog, these are defined by

$$F_a^{\mathrm{m}}(y) = \mathrm{E}\{F_a(y \mid X)\}, \qquad F_a^{\mathrm{t}}(y) = \mathrm{E}\{F_a(y \mid X) \mid A = 1\}. \tag{28}$$

The associated marginal quantile functions are obtained by inverting these standardized CDFs,

$$Q_a^{\mathrm{m}}(\tau) = \inf\{y : F_a^{\mathrm{m}}(y) \ge \tau\}, \qquad Q_a^{\mathrm{t}}(\tau) = \inf\{y : F_a^{\mathrm{t}}(y) \ge \tau\}, \tag{29}$$

while the marginal means integrate the conditional mean over the respective covariate distributions,

$$\mu_a^{\mathrm{m}} = \mathrm{E}\{\mu_a(X)\}, \qquad \mu_a^{\mathrm{t}} = \mathrm{E}\{\mu_a(X) \mid A = 1\}. \tag{30}$$

Identification of these quantities depends on the study design. In randomized experiments, randomization together with positivity implies exchangeability between treatment assignment and potential outcomes (Imbens and Rubin 2015). In observational studies, identification relies on the following conditions (Rosenbaum and Rubin 1983; Imbens and Rubin 2015).

- **Stable Unit Treatment Value Assumption (SUTVA):** there is no interference between units and no hidden versions of treatment.

- **Consistency:** if $A = a$, then the observed outcome equals the corresponding potential outcome, that is, $Y = Y(a)$.
- **Conditional ignorability:** treatment assignment is as-if randomized given covariates, so that
$$(Y(0), Y(1)) \perp A \mid X. \tag{31}$$
- **Overlap (positivity):** every covariate profile has a strictly positive probability of receiving either treatment,
$$0 < \Pr(A = 1 \mid X = x) < 1 \tag{32}$$
for all covariate values of interest.

Under these conditions, the treatment-specific conditional distributions are identified from the observed data as

$$F_a(y \mid x) = \Pr(Y \leq y \mid A = a, X = x), \qquad a \in \{0, 1\}. \tag{33}$$

In observational settings, the package employs the propensity score (PS) to support the design stage (Rosenbaum and Rubin 1983). The PS is the conditional probability of treatment assignment given covariates,

$$\rho(x) = \Pr(A = 1 \mid X = x), \tag{34}$$

and a key property of the true propensity score is that conditioning on it balances covariates across treatment groups. Following a two-stage strategy, the package first estimates $\rho(x)$ using only (A, X) and then augments the outcome model covariates with the estimated score, so that the augmented predictor vector is

$$r(x) = (x^{\mathsf{T}}, \widehat{\rho}(x))^{\mathsf{T}}. \tag{35}$$

The conditional outcome distributions are then modeled separately within each treatment arm as functions of the augmented predictor $r(x)$, and causal contrasts are computed as functionals of the estimated treatment-specific distributions. Figure 3 illustrates two treatment-specific conditional densities at a representative covariate profile. The dotted verticals mark the 90th conditional quantile for each arm, and the double-headed arrow indicates the conditional quantile treatment effect (CQTE) at that level.

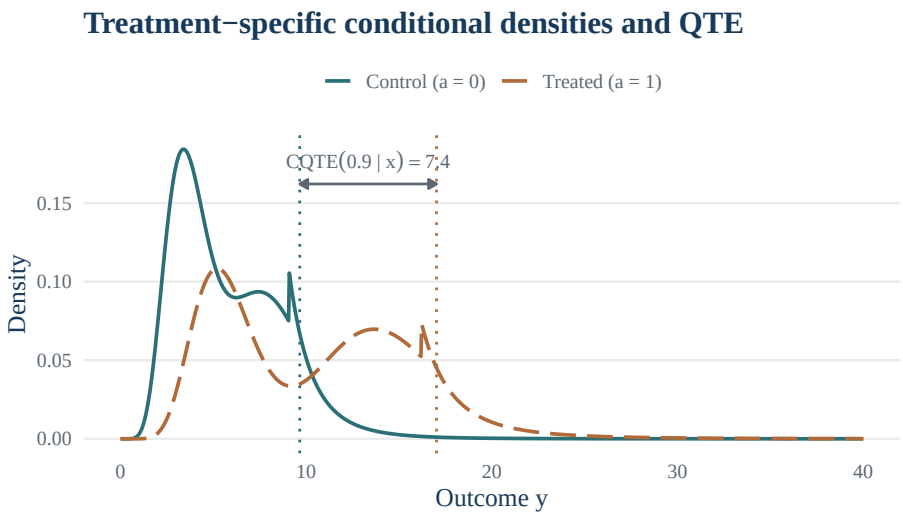


Figure 3: Treatment-specific conditional densities at a representative covariate profile. Dotted verticals mark the 90th quantile for each arm; the double-headed arrow gives the conditional quantile treatment effect (CQTE) at that level.

The package reports the causal estimands listed in Table 1, where superscripts m and t denote full-population and treated-population standardization, respectively.

Table 1: Causal estimands reported by the package

Estimand	Formula	Target population
ATE	$\mu_1^{\mathrm{m}} - \mu_0^{\mathrm{m}}$	Entire population
ATT	$\mu_1^{\mathrm{t}} - \mu_0^{\mathrm{t}}$	Treated sub-population
QTE(τ)	$Q_1^{\mathrm{m}}(\tau) - Q_0^{\mathrm{m}}(\tau)$	Entire population at index τ
QTT(τ)	$Q_1^{\mathrm{t}}(\tau) - Q_0^{\mathrm{t}}(\tau)$	Treated sub-population at index τ
CATE(x)	$\mu_1(x) - \mu_0(x)$	Individual with covariate profile x
CQTE($\tau \mid x$)	$Q_1(\tau \mid x) - Q_0(\tau \mid x)$	Individual with x at index τ

The next section describes how the package fits the treatment-specific conditional distributions and computes these estimands from posterior predictive output.

3. Package Overview: Workflow, Model, and Inference

This section describes the workflow for fitting conditional models and obtaining predictions with CausalMixGPD. The presentation begins with the one-arm setup, in which the spliced DPM–GPD model is fitted to an outcome conditional on covariates with a generalized Pareto distribution governing the upper tail above a covariate-dependent threshold. The causal extension augments this workflow with arm-specific outcome models and an optional propensity score component, reusing the same estimation and prediction machinery. Simulated data sets bundled with the package are used throughout to demonstrate the principal components; these are accessible via `data()`.

3.1. Spliced DPM–GPD model, likelihood, and priors

The observed data consist of n outcome–covariate pairs $\mathcal{D} = \{(y_i, x_i)\}_{i=1}^n$, where $x_i \in \mathbb{R}^p$ denotes the covariate vector for unit i, optionally including an intercept column.

```
data("nc_posX100_p3_k2", package = "CausalMixGPD")
dat <- data.frame(y = nc_posX100_p3_k2$y, nc_posX100_p3_k2$X)
```

The target conditional distribution $F(\cdot \mid x)$ and its spliced DPM–GPD construction are defined in Section 2, with the spliced CDF and density given in (14)–(15). For posterior computation, CausalMixGPD approximates the infinite bulk DPM by a finite truncation to J mixture components, so that

$$f_{\mathrm{DP}}(y \mid x; \Theta) \approx \sum_{j=1}^{J} w_j \, k(y \mid x; \theta_j),$$

where $j = 1, \ldots, J$ indexes components, each weight satisfies $w_j \geq 0$, and the weights sum to one up to a small residual mass assigned to the remaining components. Kernel parameters may be constant, component-specific, or covariate-dependent through link specifications of the form

$$g(\psi_j(x)) = x^\top \beta_{j,\psi}, \tag{36}$$

where $\psi_j(x)$ denotes a generic kernel parameter for component j at covariate value x and $g(\cdot)$ is a link function that enforces the parameter support (e.g., a log link for positive scale parameters). All code examples in this section use the MCMC settings shown below, which are intentionally abbreviated for illustration; longer chains and multiple chains are recommended for substantive analyses.

```
data.frame(mcmc_fixed,row.names = NULL)


  niter nburnin thin nchains seed
1  2500     500    1       3 2026
```

The one-arm spliced model is fitted by supplying a formula, data, and control arguments to `dpmgpd()`. The `formula` argument follows standard R syntax and specifies which covariates enter both the bulk and tail components. The `backend` argument selects the representation of DPM weights: `"sb"` invokes the stick-breaking representation in (11), while `"crp"` uses the Chinese restaurant process representation. The `kernel` argument specifies the bulk kernel family, `components` determines the truncation level, and `mcmc` controls the sampler configuration.

```
fit <- dpmgpd(formula = y ~ x1 + x2 + x3, data = dat, mcmc = mcmc_fixed)
```

Given the spliced density in (15), the likelihood is

$$L(\Theta, \Phi; \mathcal{D}) = \prod_{i=1}^{n} f(y_i \mid x_i; \Theta, \Phi), \tag{37}$$

where Θ collects the bulk parameters $\{w_j, \theta_j\}_{j=1}^{J}$ and Φ collects the tail parameters governing $u(x)$, $\sigma(x)$, and ξ. The log-likelihood can equivalently be written as

$$\begin{aligned}\log L(\Theta, \Phi; \mathcal{D}) \;=\; & \sum_{i=1}^{n}(1-\delta_i)\,\log f_{\mathrm{DP}}(y_i \mid x_i; \Theta) \\ & + \sum_{i=1}^{n}\delta_i\Big[\log\{1 - F_{\mathrm{DP}}(u(x_i) \mid x_i; \Theta)\} + \log f_{\mathrm{GPD}}(y_i \mid x_i; \Phi)\Big], \qquad (38)\end{aligned}$$

where f_{GPD} is the GPD tail density from Section 2.3 and $F_{\mathrm{DP}}(\cdot \mid x)$ is the bulk mixture CDF. When the GPD tail is not included, the model reduces to $f(y \mid x; \Theta, \Phi) \equiv f_{\mathrm{DP}}(y \mid x; \Theta)$, a mixture over the bulk alone. A convenience interface to this bulk-only model is provided by the `dpmix()` wrapper.

```
fit <- dpmix(formula = y ~ x1 + x2 + x3, data = dat, mcmc = mcmc_fixed)
```

Prior distributions for Θ and Φ can be specified by the user; defaults are weakly informative. For the DPM parameters Θ, the default prior employs stick-breaking priors for the mixture weights and conjugate priors for kernel parameters when conjugacy obtains. For the tail parameters Φ, the default adopts a regression-based specification for the threshold $u(x)$ and scale $\sigma(x)$, together with a weakly informative prior for the shape parameter ξ.

3.2. Inference and prediction

Posterior simulation targets the joint posterior

$$p(\Theta, \Phi \mid \mathcal{D}) \propto L(\Theta, \Phi; \mathcal{D})\, p(\Theta, \Phi)$$

using the likelihood (37) and the prior structure described in Section 3.1. CausalMixGPD conducts MCMC via nimble (R Package Authors 2025c). The fitted object supports `summary()` for posterior summaries, `plot()` for standard MCMC diagnostics, and `params()` for lightweight parameter extraction.

```
summary(fit)  # posterior summaries for parameters and functionals
params(fit)   # extract MCMC samples for parameters and functionals
plot(fit, family = "traceplot", params = "alpha") # traceplot for concentration parameter
plot(fit, family = "auto") # automatic selection of diagnostic plots for all parameters
```

The package provides posterior summaries for several functionals of the fitted conditional distribution defined in Section 2: the density $f(y \mid x)$, survival function $S(y \mid x) = 1 - F(y \mid x)$, quantile function $Q(\tau \mid x)$, and mean $\mathrm{E}(y \mid x)$. For the `density`, `survival`, and `quantile` types, each functional is evaluated draw-by-draw from (14)–(17), and the retained draws are then summarized. For `type="mean"`, the package employs analytical draw-level calculations whenever the kernel mean exists and, for spliced GPD models, the tail-shape parameter satisfies $\xi < 1$; if these conditions are not met, the ordinary mean is not reported and the user is directed to `type="rmean"`. The restricted-mean target (`type="rmean"`) provides a separate posterior predictive Monte Carlo calculation for a user-supplied cutoff and remains well-defined even when the ordinary mean does not exist. The following code illustrates how posterior density, quantile, and mean summaries are obtained at observed covariate values, together with credible intervals for quantiles and HPD intervals for means.

```
predict(fit, type = "density", interval = "credible", level = 0.95)
predict(fit, type = "quantile", index = c(0.25, 0.5, 0.75))
predict(fit, type = "mean", interval = "hpd", level = 0.9)
```

Predictions at new covariate values are obtained by calling `predict()` with a data set or covariate profiles sharing the same structure as the training data.

```
x_new <- as.data.frame(lapply(dat[, c("x1", "x2", "x3")], quantile,
                              probs = c(0.25, 0.50, 0.75), na.rm = TRUE))
y_grid <- seq(0, 10, length.out = 200)
```

For a new covariate value $x^\star$, prediction is based on the posterior predictive distribution

$$p(y^\star \mid x^\star, \mathcal{D}) = \int f(y^\star \mid x^\star; \Theta, \Phi)\, p(\Theta, \Phi \mid \mathcal{D})\, d\Theta\, d\Phi.$$

Density and survival predictions require a grid of y values supplied via the `y` argument, whereas quantile predictions require only the desired probability levels via `index`. The code below demonstrates posterior survival and quantile predictions at new covariate profiles constructed from the empirical quartiles of the training predictors.

Appendix A collects the corresponding argument-level controls. In particular, Section A.1 expands the prior and backend specifications introduced above, while Section A.2 summarizes the full prediction, interval, and diagnostic interfaces.

```
predict(fit, newdata = x_new, y = y_grid, type = "survival", level = 0.95)
predict(fit, newdata = x_new, type = "quantile", index = c(0.5, 0.99), interval = "credible")
```

3.3. Clustering extension

```
data("nc_realX100_p3_k2", package = "CausalMixGPD")
dat_cl <- data.frame(y = nc_realX100_p3_k2$y[1:100], nc_realX100_p3_k2$X[1:100, ])
```

The clustering framework reinterprets the fitted mixture as a latent partition and returns cluster summaries rather than density functionals alone. The function `dpmix.cluster()` fits the bulk-only model, and `dpmgpd.cluster()` adds the optional GPD tail for each mixture component, with predictor dependence entering through the weights, component parameters, or both. Because mixture labels are permutation-invariant, the principal outputs are label-invariant summaries: the posterior similarity matrix (PSM), a representative partition, cluster sizes, and assignment scores (Binder 1978; Dahl 2006). The same `predict()` interface handles both training labels and new-observation assignment.

A bulk-only clustering model with covariate-dependent parameters is fitted in the code below, where `type="param"` specifies that predictors enter through component-specific kernel parameters.

```
fit_cluster <- dpmix.cluster(y ~ x1 + x2 + x3, data = dat_cl[1:100,], kernel = "laplace",
                             type = "both", components = 10, mcmc = mcmc_fixed)
```

The same `predict()` interface returns either the posterior similarity matrix or cluster labels, depending on the `type` argument.

```
predict(fit_cluster, type = "psm")
```

New-observation cluster labels, together with per-cluster assignment scores, are obtained as follows.

```
predict(fit_cluster, newdata = dat_cl[101:110,], type = "label")
```

As discussed in Section 2.4, the package supports three dependence patterns and reports clustering through PSM-based summaries and representative partitions. Appendix A.3 records the clustering-specific arguments and plotting options that extend this workflow.

3.4. Causal extension and propensity score augmentation

The causal framework fits separate outcome models for each treatment arm and can optionally augment the outcome covariates with a propensity-score summary. When propensity-score augmentation is required (as in observational studies), treatment assignment is modeled as

$$A_i \mid \boldsymbol{x}_i \sim \text{Bernoulli}\big(e_{\boldsymbol{\gamma}}(\boldsymbol{x}_i)\big),$$

and the estimated propensity score is appended to the covariate vector to form the augmented predictor

$$\boldsymbol{r}(\boldsymbol{x}) = \Big(\boldsymbol{x}^\top,\ \psi(\widehat{e}(\boldsymbol{x}))\Big)^\top,$$

where the PS link function (logit or probit) is specified through the `PS` argument and the transformation $\psi(\cdot)$ applied to the estimated score is controlled by `ps_scale`, with the default being the identity. The outcome model for each arm is then a spliced DPM–GPD regression on the augmented covariates, reusing exactly the same mixture-regression and GPD-tail machinery as the one-arm workflow described in Section 3.1. For randomized experiments, the PS component is unnecessary and the same modeling workflow is applied to each arm without augmentation.

The code below loads a simulated benchmark data set and assembles the data frame expected by the causal wrappers, with columns for the outcome, treatment indicator, and covariates.

The call to `dpmgpd.causal()` fits arm-specific spliced mixture models using the same structure as `dpmgpd()`, with additional arguments controlling the propensity-score model (`PS`, `ps_scale`, `ps_summary`) and separate MCMC settings for the outcome and PS stages (`mcmc_outcome`, `mcmc_ps`).

```
causal_fit <- dpmgpd.causal(formula = y ~ x1 + x2 + x3,
                            data = causal_df, treat = "A", backend = "crp",
                            kernel = "laplace", components = 5, PS = "logit",
                            ps_scale = "logit", ps_summary = "mean",
                            mcmc_outcome = mcmc_fixed, mcmc_ps = mcmc_fixed)
```

Population-standardized estimates are obtained via the functions `ate()` and `qte()`, while treated-population-standardized versions are available through `att()` and `qtt()`. Uncertainty

quantification is provided through interval estimates, and results can be visualized by plotting the returned objects with `type="effect"` to display the estimated treatment effect curve across quantiles together with uncertainty intervals. The following code demonstrates the calculation of treated-population-standardized effects.

```
qtt <- qtt(causal_fit, probs = c(0.50, 0.90, 0.95), interval = "credible")
summary(qtt)
plot(qtt, type = "effect")
```

Conditional causal effects at user-specified covariate profiles are available through the same `predict()` interface, where the `type` argument determines whether quantile, survival, or other summaries are returned. The example below illustrates survival predictions at a grid of covariate profiles spanning the empirical quartiles of the training predictors.

```
predict(causal_fit, newdata = causal_xgrid, type = "survival", y = rep(4, nrow(causal_xgrid)))
```

The bulk-only analogue is `dpmix.causal()`, which retains the same causal estimand and prediction interface without the GPD tail. For randomized experiments, the same workflow applies without PS augmentation, and the `PS` argument can be set to `NULL`. Additional causal estimands, propensity-score controls, and plotting options are catalogued in Appendix A.3.

4. Data Analysis I: Clustering

In this section, the procedure is explained for conducting clustering in the context of a conditional mixture model for the case of a bulk-only setting. The goal is to discover clusters amongst the observational units through the use of the fitted mixture model, while simultaneously accounting for uncertainty in the structure of such clusters in a label invariant way. While previously we focused on illustrating how prediction can be conducted through predictive densities in these models, here our interest lies in partition summaries and predicting cluster memberships in out-of-sample data. Important results include the posterior similarity matrix (PSM), a summary of partitions, and cluster memberships of out-of-sample data.

4.1. Data and preprocessing

The Boston data set is used in this analysis, where each observation corresponds to a census tract. The response variable used is medv, representing median home prices in each area. Predictors used include the lstat, rm, and nox variables, which represent the level of neighborhood disadvantage, the size of the houses, and pollution levels, respectively. This combination of predictors represents a sensible choice for discovering representative clusters. We are interested in uncovering representative clusters from amongst the data, and predicting which latent clusters held-out census tracts belong to using out-of-sample prediction. A 80%/20% train-test split was conducted.

```
library(CausalMixGPD)
library(MASS)

set.seed(123)
data("Boston", package = "MASS")
```

```
dat <- Boston
dat <- subset(dat, select = c(medv, lstat, rm, nox))

n <- nrow(dat)
idx_train <- sample(seq_len(n), size = floor(0.80 * n), replace = FALSE)
train_dat <- dat[idx_train, ]
test_dat  <- dat[-idx_train, ]
```

In Figure 4, we show the descriptive analyses of the Boston housing price data as represented by the median home value (`medv`). The variability in the response variable is quite large, and there seems to be an elongated tail toward higher values. The implication of these observations is that there would be a more accurate model of structure when using a flexible mixture approach, rather than a simple regression.

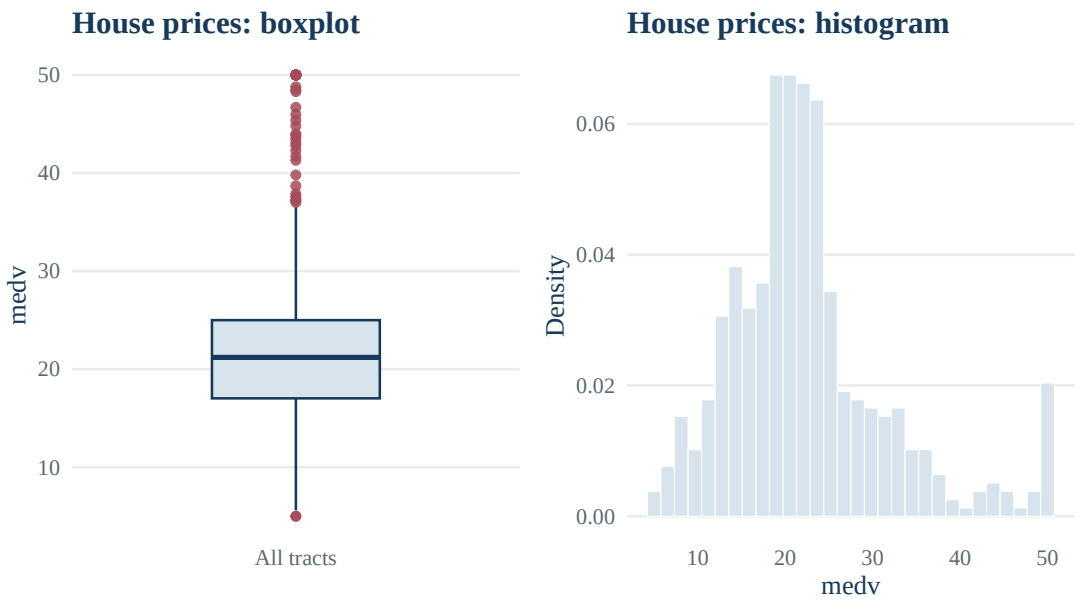


Figure 4: Full-sample distribution of Boston house prices (`medv`): boxplot (left) and histogram (right).

4.2. Model estimation and training cluster summaries

A bulk only conditional mixture model is applied to the `medv` dataset with three covariates. When `type="weights"`, covariates determine the weights of the mixture while the parameters of the components are common across all covariates. This method is helpful for clustering where the occurrence rate of hidden clusters may differ between neighborhoods.

```
fit_clust <- dpmix.cluster( formula = medv ~ lstat + rm + nox,
                            data  = train_dat, kernel   = "normal",
                             type = "weights", components = 10, mcmc = mcmc_fixed)
```

Since mixture components cannot be identified up to permutation, clusters in the training data are represented using the posterior similarity matrix and sample training labels. The posterior similarity matrix gives the main label-invariant uncertainty summary via its storage of posterior co-clustering probabilities.

```
z_train_psm <- predict(fit_clust, type = "psm")
```

```
plot(z_train_psm, type = "summary")
```

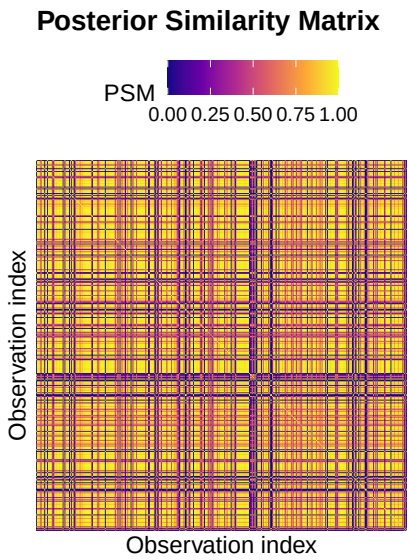


Figure 5: Posterior similarity matrix summary for the Boston training sample.

As seen from Figure 5, the PSM reveals clear block structure along the diagonal, implying that the model can detect multiple sets of tracts whose co-clustering possibility is relatively high. While there are clear separations, there are still some uncertainties for those tracts on the edge of two or more neighboring groups, which reflects reasonable posterior uncertainty in a non-experimental housing dataset, since features of neighborhoods tend to change gradually rather than abruptly.

```
z_train_lab <- predict(fit_clust, type = "label")
```

Figure 6 shows the representative training cluster summary produced by the package `plot()` method applied to the training labels object.

```
plot(z_train_lab, type = "summary")
```

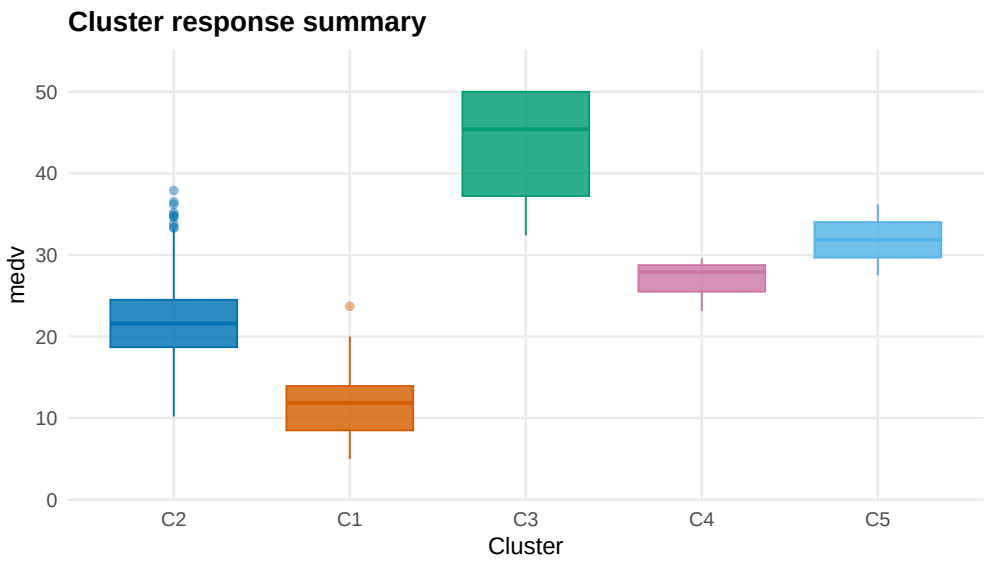


Figure 6: Representative training-cluster summary for the Boston `medv` analysis.

The representative partition translates the posterior co-clustering structure into an interpretable summary on the response scale. The inferred groups separate tracts primarily by housing-value level, but they also reflect broader differences in neighborhood composition and environmental quality. Thus, the clustering output is not merely a numerical partition induced by the mixture fit; it yields a meaningful summary of heterogeneity in the Boston housing data.

4.3. Prediction for new observations

Cluster assignment for new observations is performed through the same prediction interface. The output is a label in the space of the representative training clusters, obtained by aggregating posterior predictive co-clustering evidence at the cluster level.

```
z_test <- predict( fit_clust, newdata = test_dat, type    = "label")
```

Figure 7 displays the distribution of predicted cluster assignments across the test sample, produced by the package `plot()` method applied to the test labels object.

```
summary(z_test)$cluster_profiles


  cluster  n medv_mean  medv_sd lstat_mean lstat_sd
1      C4 53  21.57736 6.867192  13.415660 5.602698
2      C2 24  18.69583 5.343951  14.474583 7.229240
3      C5 18  35.15556 9.010988   5.679444 1.503887
4      C1  7  11.72857 2.293261  24.204286 6.162083
   rm_mean     rm_sd  nox_mean     nox_sd
1 6.225340 0.6417186 0.5775660 0.11813723
2 5.976583 0.4887950 0.5618333 0.12035912
3 7.260167 0.6143898 0.4957222 0.09672649
4 5.763571 0.5772723 0.6538571 0.09705227
```

```
plot(z_test, type = "sizes")
```

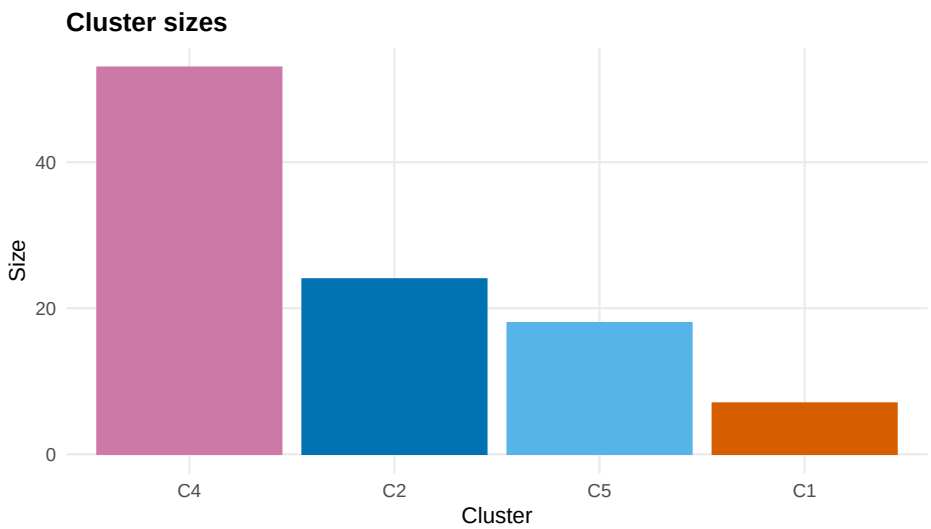


Figure 7: Predicted cluster assignment distribution for the test sample (`medv` data). Each bar corresponds to one representative training cluster; bar heights show the number of test observations assigned to that cluster.

The expected characteristics of the clusters' profiles are consistent with the interpretation of the representative partition. Cluster C5 contains the wealthiest tracts defined by a high mean house value, low average `lstat`, large mean number of rooms, and relatively low pollution. Cluster C1 represents the poorest neighborhoods with a low average house value, significant disadvantages in terms of neighborhood quality, low mean number of rooms, and a high level of pollution. Clusters C2 and C4 represent intermediate cases which show that multiple clusters were detected within the housing market rather than a bimodal distribution with poor and wealthy neighborhoods at its ends.

Predictably, the vast majority of held-out tracts are classified into middle-value clusters (C4 and C2) and there is no tendency for some tract assignments to the cluster representing the low end. This makes sense for the Boston data and confirms that the predictions are not overly affected by the presence of the outliers in the dataset. More significantly, tract assignments in the test set validate the sensible interpretation of the training set partition: tracts that belong to more expensive clusters are characterized by less neighborhood disadvantage, larger homes, and a cleaner environment; while tracts associated with poorer clusters possess the opposite characteristics.

This case study provides a perfect illustration of our workflow where the inference was provided in terms of a representative partition and cluster descriptions while the predictive method assigned new observations into those clusters using the probabilistic criteria for membership into the same latent class in the posterior predictive distribution. Representative clusters become even more helpful in this setting because they provide information about the link between clustering uncertainty and the distribution of the response and covariates. In this example, the latent clusters are consistent with the social and environmental gradient

quite well which means that the nonparametric mixture model can capture stable patterns despite the heterogeneity of the response distribution. We are moving on to causal comparisons (Section 5) at this point.

5. Data Analysis II: Causal Inference

In this case study, the estimands corresponding to the two-arm spliced distributional regression are described. In contrast to the partition estimation considered in Section 4, here, interest lies in mean and quantile differences in terms of both unconditional and conditional causal estimands.

Data and estimands

In order to demonstrate the estimation of unconditional and conditional causal estimands using the proposed approach, the `lalonde` dataset from the MatchIt R package is used for benchmarking purposes. Here, the outcome is the 1978 annual income level measured by `re78` ($), and the treatment indicator equals to the indicator for program participation (variable `treat`). The set of covariates includes age, years of schooling, race, marriage status, college degree status, together with previous earning variables: `re74` and `re75`. It should be emphasized that earnings typically possess considerable skewness, cluster around zero values and have a strong upper tail, which gives motivation for the consideration of distributional summaries instead of mean only.

Estimation of the four causal estimands described in Table 1 is performed based on Section 2.5. Standardized versions of causal estimands (ATE, QTE) are averaged across the empirical distribution of $\{\boldsymbol{x}_i\}_{i=1}^n$, while conditional versions (CATE, CQTE) are estimated for certain individuals.

At first, the required data are loaded, and the design matrix is built for further estimation of causal estimands.

```
data("lalonde", package = "MatchIt")
app_lalonde_covars <- within(lalonde, {
  race <- factor(race, levels = c("white", "black", "hispan"))
  married <- factor(married, levels = c(0, 1), labels = c("no", "yes"))
  nodegree <- factor(nodegree, levels = c(0, 1), labels = c("no", "yes"))
})
app_lalonde_scale_vars <- c("age", "educ", "re74", "re75")
app_lalonde_scale_center <- vapply(
  app_lalonde_scale_vars,
  function(v) mean(app_lalonde_covars[[v]], na.rm = TRUE),
  numeric(1)
)
app_lalonde_scale_scale <- vapply(
  app_lalonde_scale_vars,
  function(v) stats::sd(app_lalonde_covars[[v]], na.rm = TRUE),
  numeric(1)
)
for (v in app_lalonde_scale_vars) {
  app_lalonde_covars[[paste0(v, "_z")]] <-
    (app_lalonde_covars[[v]] - app_lalonde_scale_center[[v]]) /
    app_lalonde_scale_scale[[v]]
}
app_lalonde_x_formula <- ~ age_z + educ_z + race + married + nodegree + re74_z + re75_z
```

```
app_lalonde_y <- (app_lalonde_covars$re78 + 0.5) / 1000
app_lalonde_A <- as.integer(app_lalonde_covars$treat)
app_lalonde_taus <- c(0.25, 0.50, 0.75, 0.90, 0.95)
app_lalonde_X <- stats::model.matrix(app_lalonde_x_formula, data = app_lalonde_covars
                                                )[, -1, drop = FALSE]
```

Figure 8 provides full-sample descriptive views of observed 1978 earnings (`re78`) by treatment assignment. Both groups exhibit substantial mass near zero together with pronounced right-skewness, and the arm-specific distributions differ in ways that are not well summarized by a simple location shift. These features motivate modeling the full treatment-specific outcome distributions directly rather than relying solely on average effects.

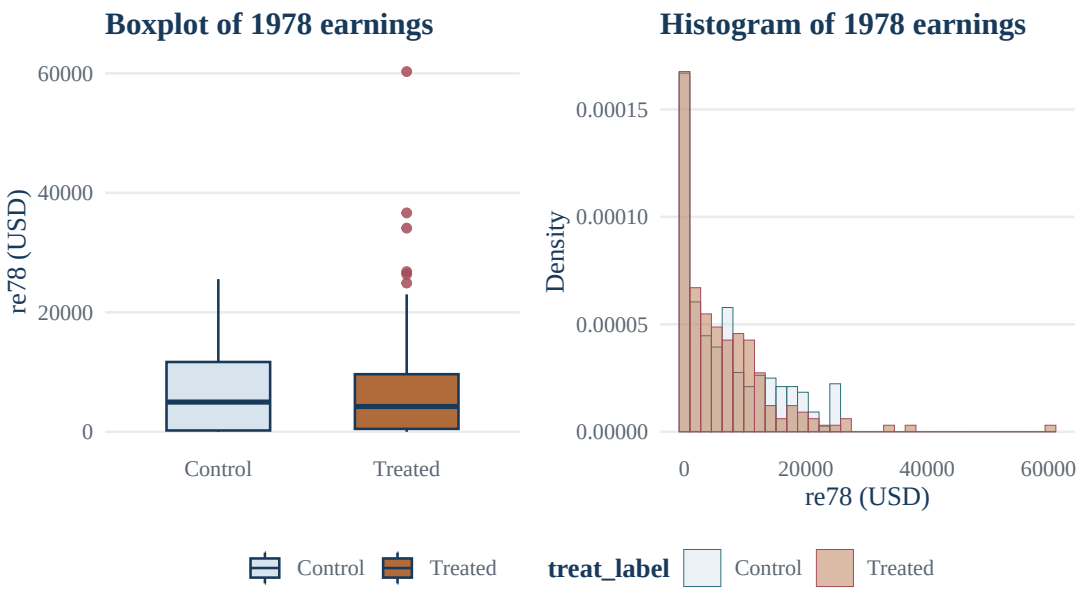


Figure 8: Full-sample distribution of Lalonde 1978 earnings (`re78`): boxplot by treatment arm (left) and overlaid histograms by treatment arm (right).

5.1. Model and fit

Arm-specific conditional outcome distributions $F_a(\cdot \mid \boldsymbol{x})$ are modeled via the spliced bulk–tail construction described in Section 2.3:

$$f_a(y \mid \boldsymbol{x}) = \sum_j w_{aj}(\boldsymbol{x})\, k(y \mid \Theta_{aj})\, \mathbf{1}(y \le u) + \{1 - F_a(u \mid \boldsymbol{x})\}\, g(y \mid u, \sigma_a, \xi_a)\, \mathbf{1}(y > u).$$

The bulk kernel $k(\cdot)$ represents a Dirichlet process mixture regression with a Chinese restaurant process backbone, while the tail density $g(\cdot)$ follows the GPD distribution with positive scale $\sigma_a > 0$ and tail index ξ_a. In our model, the GPD tail index ξ_a regulates extremal weight (e.g., if $\xi_a > 0$, it indicates a heavier tail). Our model mitigates estimation instabilities for extreme quantiles through nonparametric bulk and EVT-consistent tail modeling.

Here, a two-sample mixture is fitted with the gamma bulk kernel and the same MCMC configuration across the analysis. The aim of this analysis is not to make any final conclusions but to highlight how the package provides standardized and conditional Lalonde earnings contrasts under the spliced DPM-GPD framework.

```
fit <- dpmgpd.causal(y = app_lalonde_y, X = app_lalonde_X, treat = app_lalonde_A,
                     backend = "crp", kernel = "gamma", components = 10, mcmc = mcmc_fixed)
```

5.2. Posterior summaries

Before reporting treatment contrasts, it is useful to view the fitted causal object as defining two arm-specific outcome distributions from which all reported summaries are derived. The value of the framework is therefore not limited to a single regression coefficient or scalar contrast; instead, mean, quantile, and tail-sensitive causal estimands are all obtained as coherent functionals of the same fitted posterior distribution.

```
  total    ps   con   trt parallel_arms
1 356.3 36.71 211.3 108.2         FALSE
```

5.3. Standardized estimands: ATE and QTE

The overall treatment effect is summarized with ATE and QTE.

```
ate_fit <- ate(fit, interval = "hpd", level = 0.95)
summary(ate_fit)

ATE Summary
==================================================
Prediction points: 1
Conditional: NO | PS used: YES
Interval: hpd

Model specification:
  Backend (trt/con): crp / crp
  Kernel (trt/con): gamma / gamma
  GPD tail (trt/con): YES / YES

ATE estimates:
 estimate  lower upper
   -0.766 -2.992 1.358
```

The posterior ATE estimate is negative on the scaled earnings scale, but the corresponding interval includes zero. Thus, after averaging over the empirical covariate distribution, the analysis does not provide strong evidence of a uniform overall mean effect of the training program. This is important because it shows that mean-based summaries alone do not yield a decisive conclusion in this application.

```
qte_fit <- qte(fit, probs = c(0.25, 0.50, 0.75), interval = "credible")
```

Figure 9 displays two standardized QTE views side by side from the spliced causal model: the effect curve and the treated-versus-control quantile curves with uncertainty intervals.

```
qte_effect_plot <- plot(qte_fit, type = "effect")
qte_arms_plot <- plot(qte_fit, type = "arms")
qte_effect_plot + qte_arms_plot + plot_layout(ncol = 2)
```

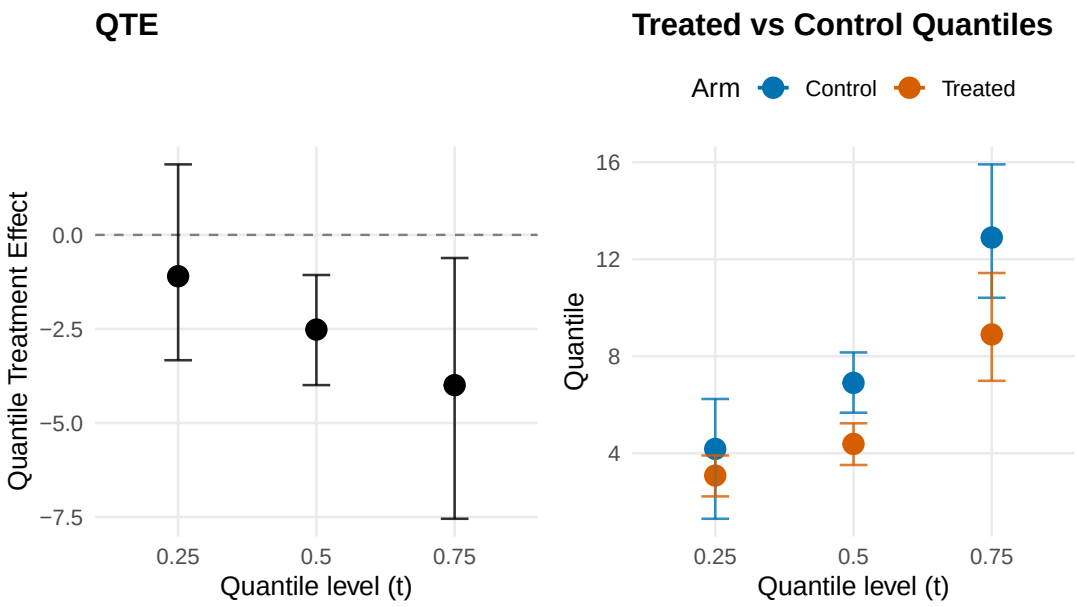


Figure 9: Standardized QTE visual summaries for Lalonde 1978 earnings: effect curve (left) and treated-versus-control quantiles (right), each with pointwise credible intervals.

Nevertheless, when considering the comparison between the quantiles for the two distributions, it can be noted that the QTEs at the specified quantiles are still negative, which implies that the treatment distribution stands below the control one along the considered parts of the marginal distribution. At the same time, the consideration of the quantile curves of both distributions shows that their deviations cannot be explained just by an average shift but also reflect the heterogeneity of their distribution.

5.4. Conditional estimands for representative profiles

In order to show the heterogeneity of the treatment effect, the conditional estimand is calculated for four covariate profiles representing the whole set of the values of the covariates. The covariate values in the profiles are presented on the actual data scale thanks to the use of sample quantiles for continuous variables and the values observed for categorical covariates. The initial values of the profiles are centered and scaled according to the values of the training set.

Table 2: Representative Lalonde participant profiles used for conditional causal contrasts

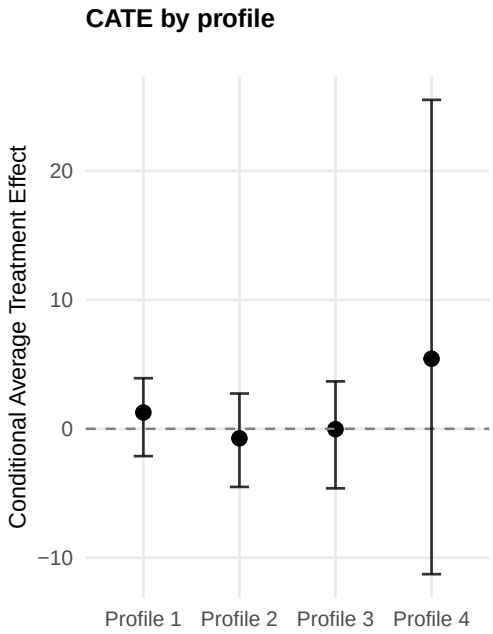
	Profile 1	Profile 2	Profile 3	Profile 4
age	20.0000	25.0000	32.0000	43.0000
educ	9.0000	11.0000	12.0000	13.0000
race	black	hispan	white	black
married	no	no	yes	yes
nodegree	yes	no	no	yes
re74	0.0000	47.4143	2705.7476	9383.2356
re75	0.0000	16.4710	1389.6486	4201.8870

```
cate_fit <- cate(fit, newdata = xnew, interval = "hpd", show_progress = FALSE)
summary(cate_fit)
# plot(cate_fit, type = "effect") # Displays mean-based CATE plot by profile
```

The mean-based conditional average treatment effects are generally modest and uncertain. Although some profiles show positive point estimates, the corresponding intervals are broad and typically overlap zero. Thus, at the level of conditional means, the program does not appear to induce a strong and uniform shift in earnings across all participant types.

```
cqte_fit <- cqte(fit, probs = c(0.25, 0.50, 0.75, 0.90, 0.95),
                 newdata = xnew, interval = "credible")
# cqte_fit$fit_df # Displays the full CQTE posterior summary data frame
```

Figure 10 places the mean-based CATE plot beside a compact CQTE table so the profile-specific mean and quantile contrasts can be read together. The table entries report posterior estimates with 95% credible intervals in brackets.



CQTE with 95% credible intervals

	P1	P2	P3	P4
0.25	2.07 [−0.00, 3.55]	0.26 [−4.34, 3.39]	0.08 [−4.60, 3.42]	2.15 [−0.00, 3.66]
0.50	1.67 [−2.23, 4.12]	−2.02 [−5.52, 2.22]	−1.61 [−5.38, 1.24]	3.05 [−4.20, 13.28]
0.75	0.44 [−3.87, 5.38]	−2.07 [−6.01, 2.05]	−1.03 [−5.94, 3.15]	4.82 [−14.98, 32.88]
0.90	0.21 [−6.23, 7.18]	−1.24 [−6.93, 5.97]	0.50 [−5.67, 7.96]	8.71 [−30.39, 65.45]
0.95	0.79 [−8.46, 10.88]	−0.31 [−8.23, 11.01]	2.31 [−6.63, 14.03]	14.82 [−41.45, 104.81]

Figure 10: Mean-based CATE estimates (left) and compact CQTE summaries (right) for Lalonde earnings at the four representative profiles. CQTE entries report posterior estimates with 95% credible intervals.

CQTE analysis implies heterogeneity of treatment effects by both profiles and quantile level. On several occasions, the effect of the program for low- and median-quantile contrasts appears to be close to zero or even negative. At the same time, uncertainty about treatment effects is high at the upper quantiles, which is not surprising since upper-quantile contrasts include very few observations and thus are more sensitive to the spliced tail. Overall, the analysis reveals no evidence of any positive effects of the program but only suggests that, if any positive effects occur, they can be profile-specific.

5.5. Results and interpretation

Two main questions that have to be answered in the interpretation of the results concern the general effects of the program (the question of interest here) on the earnings distribution, as well as the differences between effects for different profiles of participants. In this regard, the answer to the first question does not provide anything significant. The average treatment effect is close to zero, and the estimated QTE are negative at lower and mid-quantiles. On the other hand, the second question is answered in somewhat mixed manner – treatment effects are profile-specific, and some CQTE at upper quantiles show a positive effect which seems to be true only for some particular subpopulations.

In general, this analysis emphasizes the importance of using distributional analysis in contexts of heavy-tailed data. A simple analysis that concerns only the mean will provide little evidence of a program effect. Meanwhile, QTE summary allows us to observe the heterogeneity of treatment effects.

6. Discussion

CausalMixGPD integrates flexible bulk density estimation, optional EVT-based tail extrapolation, predictor-dependent clustering, and distributional causal inference within a single coherent workflow. The same fitted model supports posterior summaries for densities, survival curves, analytical means (when they exist), restricted means, quantiles, and treatment contrasts, thereby obviating the fragmented analysis pipelines often required when heavy tails and distributional effects are both of concern.

The package is considerably broader than the spliced examples emphasized in this article. Users may work with bulk-only or spliced models, multiple kernel families on the real line or on positive support, stick-breaking or CRP backends, and clustering or causal wrappers built upon the same predictive machinery. This shared infrastructure facilitates transitions from exploratory subgroup analysis to tail-focused prediction and causal reporting without altering the underlying modeling language.

Several practical limitations warrant acknowledgment. First, computation can be demanding. Mixture models with covariate dependence, optional tail components, and arm-specific causal fits are inherently MCMC-intensive, and costs escalate further with sensitivity analyses, longer chains, or richer predictor sets.

Second, model tuning requires attention. A spliced analysis inherits specification choices from both the DPM and peaks-over-threshold components, including kernel family, truncation level, threshold specification, and the number of exceedances available for tail learning. When overlap is weak or exceedances are sparse, posterior uncertainty can be substantial; for clustering, the posterior similarity matrix is often more informative than any single partition summary.

Third, the current scope is circumscribed. The causal implementation is restricted to binary treatments and continuous outcomes under standard consistency, ignorability, and overlap assumptions. Propensity-score augmentation can improve covariate adjustment in observational studies, but it is not a substitute for careful study design, overlap diagnostics, or sensitivity analysis for unmeasured confounding.

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A. Customization of Package and Additional Features

This appendix details the customizable features of the package which are not mentioned in the main part. The objective is to provide a concise guide on customizing the package in terms of model fitting process, manipulating posterior results, customization in specific application case, and other useful functions of CausalMixGPD.

A.1. Model fitting options and prior specifications

Besides the customizable features illustrated in Sections 2.1, 2.2, 2.5, and 3.1, this section provides a brief introduction to arguments that allow kernel choice, latent structure, and prior customization.

Kernel families and support

The argument `kernel` allows to choose among seven different kernel families, which are either defined with positive support or the real line. The kernels with positive support are `"gamma"`, `"lognormal"`, `"invgauss"`, and `"amoroso"`. These four kernels provide the respective distribution functions needed for mixture model workflow. The remaining three kernels with the real support are `"normal"`, `"laplace"`, and `"cauchy"`. Because the moments for these three kernels do not exist, they cannot be used for GPD splicing. Function `kernel_support_table()` gives information of the support, whether the kernel is compatible with GPD or not, the support of the covariates, and the support of the backend. The function `get_kernel_registry()` gives the kernel registry with detailed information such as the parameter names, the default mode of prior, and the density/CDF/quantile functions.

Backend type and truncation rule

The argument `backend` determines the structure of the model for mixing coefficients. The argument `sb` stands for the truncated stick-breaking scheme while `crp` denotes CRP latent allocation scheme. The latter is a variant of the former with argument `GPD=TRUE`. This backend allows for thresholds, tail scales, and tail shapes per mixture component. For causal models, the argument `backend` can be either a string of one kind or a vector of length two because different causal arms can have different latent allocation schemes. Arguments `components` and `epsilon` give the maximum number of mixture components (default to be the size of the sample) and the threshold, respectively. Note that these two arguments can be customized separately for each causal arm.

Parameter specification and prior setting

The argument `param_specs` indicates a named list giving the parameters of the kernel or tail distribution. The element of this argument includes kernel/tail parameter specifications. Kernel/tail parameter has three types of modes, namely, `"fixed"`, `"dist"`, and `"link"`. In the case of using `dist`, prior can adopt normal, gamma, inverse-gamma, or log-normal distribution. Alternatively, with mode being `"link"`, the parameter of kernel/tail will be related through a linear regression with the covariates based on the five types of link functions, that is, `"identity"`, `"exp"`, `"log"`, `"softplus"`, and `"power"`. Here the link power will be `link_power`. Kernel/tail parameter specification is provided in the default registry. The arguments with the same structure can be used for thresholds, tail scales, and tail shapes,

which can take some fixed values where necessary.

A.2. Fitting, prediction, and posterior output

This section gathers the arguments used after a model class has been chosen: sampler configuration, monitoring, prediction targets, interval summaries, and extraction utilities.

MCMC configuration. `mcmc` controls `niter`, `nburnin`, `thin`, `nchains`, and `seed`; causal models split this into `mcmc_outcome` and `mcmc_ps`. Runtime controls include `show_progress`, `quiet`, `parallel_chains` with `workers`, `parallel_arms` for arm-wise causal fitting, and `timing`. CRP-based backends additionally use `z_update_every` to tune the frequency of global label updates.

Monitoring and fitting interfaces. `monitor="core"` retains the main structural parameters, whereas `"full"` retains all nodes. `monitor_latent=TRUE` stores allocation labels $z_1, \ldots, z_n$, and `monitor_v=TRUE` stores stick-breaking fractions for SB models. Models can be supplied through a formula interface (`formula`, `data`) or directly through `y` and `X`; causal fits additionally use `treat`. The `bundle()` plus `mcmc()` workflow separates model construction from sampling when the generated nimble object needs to be inspected or modified first.

Prediction types and intervals. `predict()` supports `type="density"`, `"survival"`, `"quantile"`, `"mean"`, `"rmean"`, `"median"`, `"sample"`, and `"fit"`. These return posterior summaries of $f(y \mid x)$, $S(y \mid x)$, $Q(\tau \mid x)$, $\mathrm{E}(Y \mid x)$, the restricted mean $\mathrm{E}[\min(Y, c) \mid x]$ via `cutoff` and `nsim_mean`, the median (as a special case of the quantile method), posterior predictive draws via `nsim`, or retained draw-level fitted values for custom checks. Interval summaries are controlled by `interval="credible"`, `"hpd"`, or `NULL`, together with `level` and `probs`.

Summary and extraction methods. `summary()` reports posterior means, standard deviations, quantiles, and ESS, with `pars` and `ps_pars` available for filtering. `params()` reshapes posterior means into component- and coefficient-level layouts; `fitted()` and `residuals()` provide observation-level predictive summaries; and `ess_summary()` adds per-chain, time-normalized, and robust ESS diagnostics.

A.3. Workflow-specific extensions

This section records controls specific to the clustering and causal workflows in Sections 3.3, 3.4, 4, and 5. Representative outputs appear in Figures 6, 7, 9, and 10, with the latter combining the CATE plot and CQTE table.

Causal estimands. Beyond `ate()`, `qte()`, `cate()`, and `cqte()`, the package provides `att()` for effects averaged over the treated covariate distribution, `qtt()` for quantile effects on the treated, and `ate_rmean()` for restricted-mean contrasts. The `ate()` and `cate()` families also accept `type="mean"` or `"rmean"` with a corresponding `cutoff`. All effect objects support `summary()`, `print()`, and `plot()`.

Propensity score customization. Observational causal fits can estimate propensity scores with `PS="logit"`, `"probit"`, `"naive"`, or `FALSE`. The estimated score is appended to the outcome model on either the logit scale (`ps_scale="logit"`) or probability scale (`"prob"`); posterior summaries use `ps_summary="mean"` or `"median"`. Numerical stability is controlled through `ps_clamp`, coefficient shrinkage through `ps_prior`, and the two-stage fitting scheme through separate `mcmc_ps` and `mcmc_outcome` lists.

Clustering workflow and prediction types. `dpmix.cluster()` and `dpmgpd.cluster()` accept `type="weights"`, `"param"`, or `"both"` to place predictor dependence in the mixture weights, kernel parameters, or both. Their `predict()` methods return either a posterior similarity matrix (`type="psm"`) or representative labels under the Dahl criterion (`type="label"`), including assignments for `newdata`. The associated `summary()` method reports cluster sizes and within-cluster covariate/response profiles.

Visualization and extraction. `plot()` for one-arm fits covers standard MCMC diagnostics, including `"traceplot"`, `"density"`, `"histogram"`, `"running"`, `"autocorrelation"`, `"crosscorrelation"`, `"Rhat"`/`"grb"`, `"effective"`, `"geweke"`, `"caterpillar"`, `"pairs"`, and `family="auto"`. Causal plots route by arm and offer `type="effect"`, `"arms"`, or `"both"`; QTE plots additionally use `facet_by`; cluster plots provide PSM heatmaps plus label summaries via `type="sizes"`, `"certainty"`, and `"summary"`. All plots return ggplot2 objects and can be converted to plotly. Companion extraction tools include `summary()`, `params()`, `fitted()`, `residuals()`, and `ess_summary()`.

A.4. Utilities and computing support

The third and last section includes a selection of helper functions and resources that accompany the main modeling process.

Distributional utilities. The package exposes the complete R-style `d/p/q/r` distributions for generalized Pareto distribution, every supported kernel mixture, and every spliced kernel mixture and GPD distribution; typical examples include `dgpd()`, `dgammamix()`, and `dgammamixgpd`. They are particularly useful for prior checking, generating simulations, benchmarking, and other purposes requiring distributional manipulations outside fitted models.

Simulators and bundled datasets. Functions `sim_bulk_tail()`, `sim_causal_qte()`, and `sim_survival_tail()` provide data generation for reproducible benchmarks for bulk and tail and causal analyses; random number generation seeds are supplied by the `seed` argument. The package comes bundled with twelve datasets with unconditional/conditional/causal, positive/real-line, and tail/no-tail designs, each featuring outcomes, covariates, treatment indicators where relevant, metadata, and the true values of interest.

Computational considerations. Large-scale analyses allow the use of `parallel_chains` with `workers`, `parallel_arms` for causal fits, `chunk_size` for predicting in parallel chunks, and `ndraws_pred` for sampling from the posterior predictive distribution of parameters. The `show_progress` and `quiet` arguments can be used to control progress reporting and compiler

verbosity, respectively, while `timing=TRUE` saves time spent on each fit for benchmarking purposes.

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